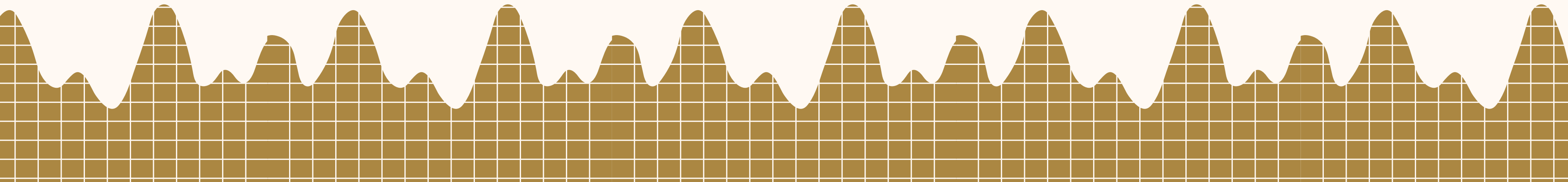
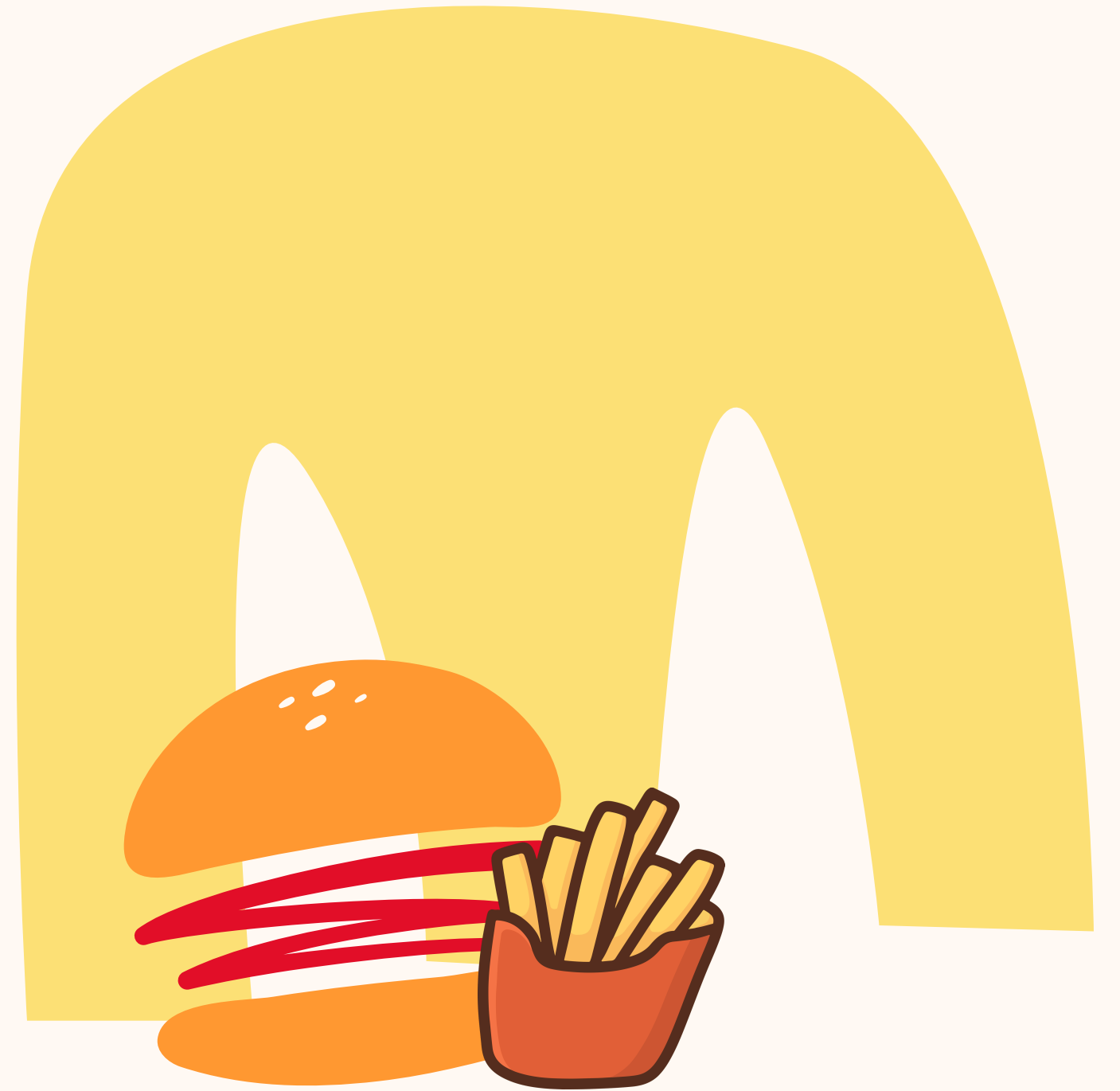


McDonald's analysis

McDonald's Branch Profitability Analytics
Platform

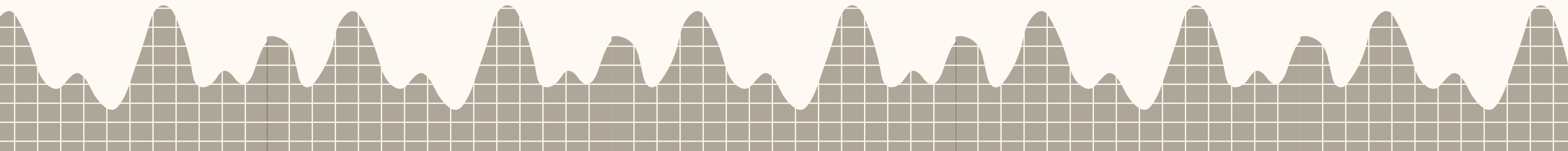
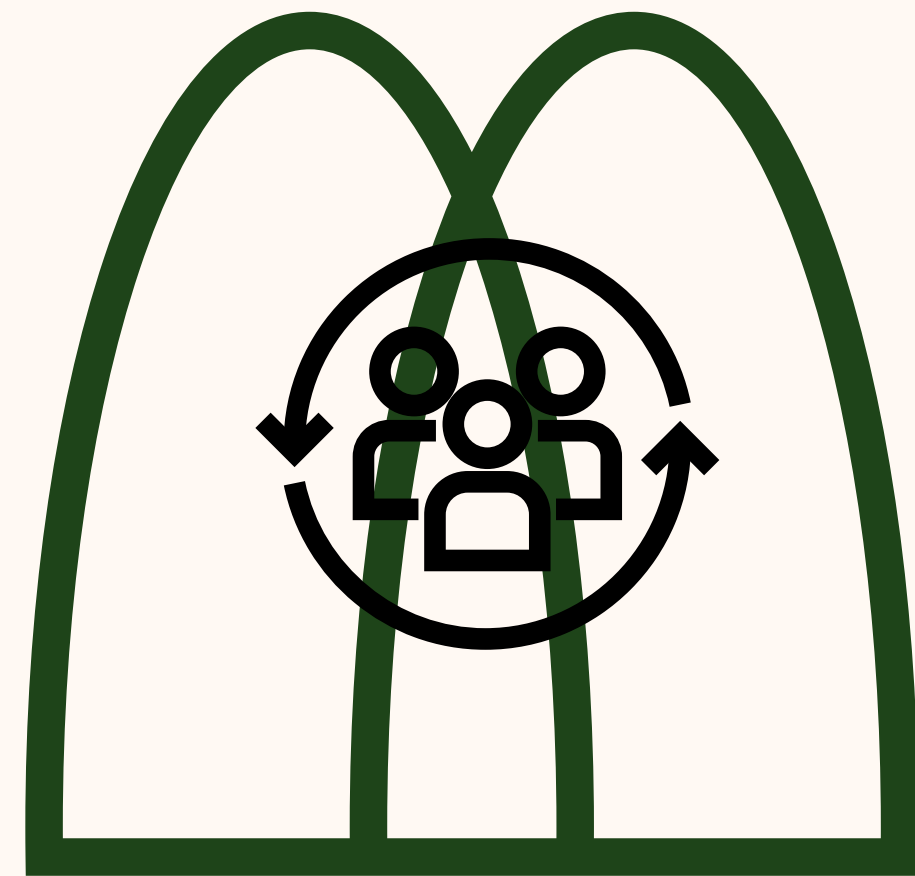




Business Context

about McDonald's

McDonald's is one of the world's largest fast-food chains, operating thousands of branches globally and serving millions of customers daily. Its business relies heavily on operational efficiency, customer satisfaction, and consistent service quality across locations.





Business Context

Business Challenge

Branch profitability is influenced by multiple factors, including:

- Customer experience and satisfaction
- Delivery performance and service speed
- Operational efficiency
- Geographic and branch characteristics

Identifying how these factors interact is difficult due to the large volume, velocity, and variety of data generated across branches.

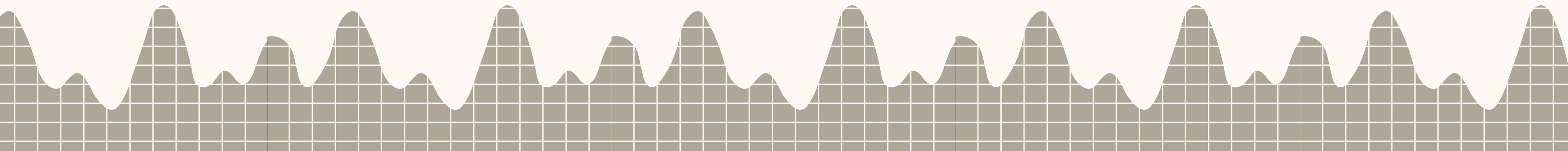




Business Context

Project Goal

Develop a modern data platform capable of integrating operational, customer, and store data to identify profitability drivers and predict underperforming branches.





Business Question

Business Question

How can McDonald's identify factors that negatively impact branch profitability and predict underperforming branches?

Stakeholders:

store managers

Focus: Daily branch performance and operational improvements.

Regional Managers

Focus: Comparing performance across multiple branches and regions.

Operations Teams

Focus: Improving service efficiency and delivery performance.

Data Teams

Focus: Generating insights and predictive models.





Data Sources Overview

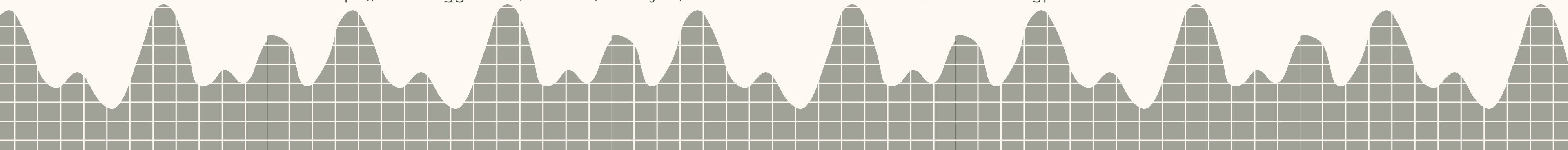
Source Type	Dataset	Business Purpose	Main Attributes	Integration Role
Streaming Source	Food Delivery Dataset	Monitor operational performance and delivery efficiency in real-time	order_id, delivery_time, order_time, distance, traffic_level, delivery_person_age, vehicle_condition	Provides operational events used to measure service quality, delays, and efficiency metrics
Late Arrival Source	McDonald's Store Reviews Dataset	Measure customer experience and delayed feedback impacting profitability	store_name, rating, review_text, review_time, store_address	Customer reviews may arrive hours or days after the customer interaction, requiring late-arrival handling
Batch / Static Source	McDonald's Stores Dataset	Provide store metadata and branch context for enrichment and segmentation	store_name, city, country, services, latitude, longitude	Used as reference data for geographic analysis, branch enrichment, and static dimensions enrichment

Datasets:

Food Delivery: https://www.kaggle.com/datasets/varshinipallerla/food-delivery?utm_source=chatgpt.com

McDonald's Store Reviews: https://www.kaggle.com/datasets/nelgiriyeewithana/mcdonalds-store-reviews?utm_source=chatgpt.com

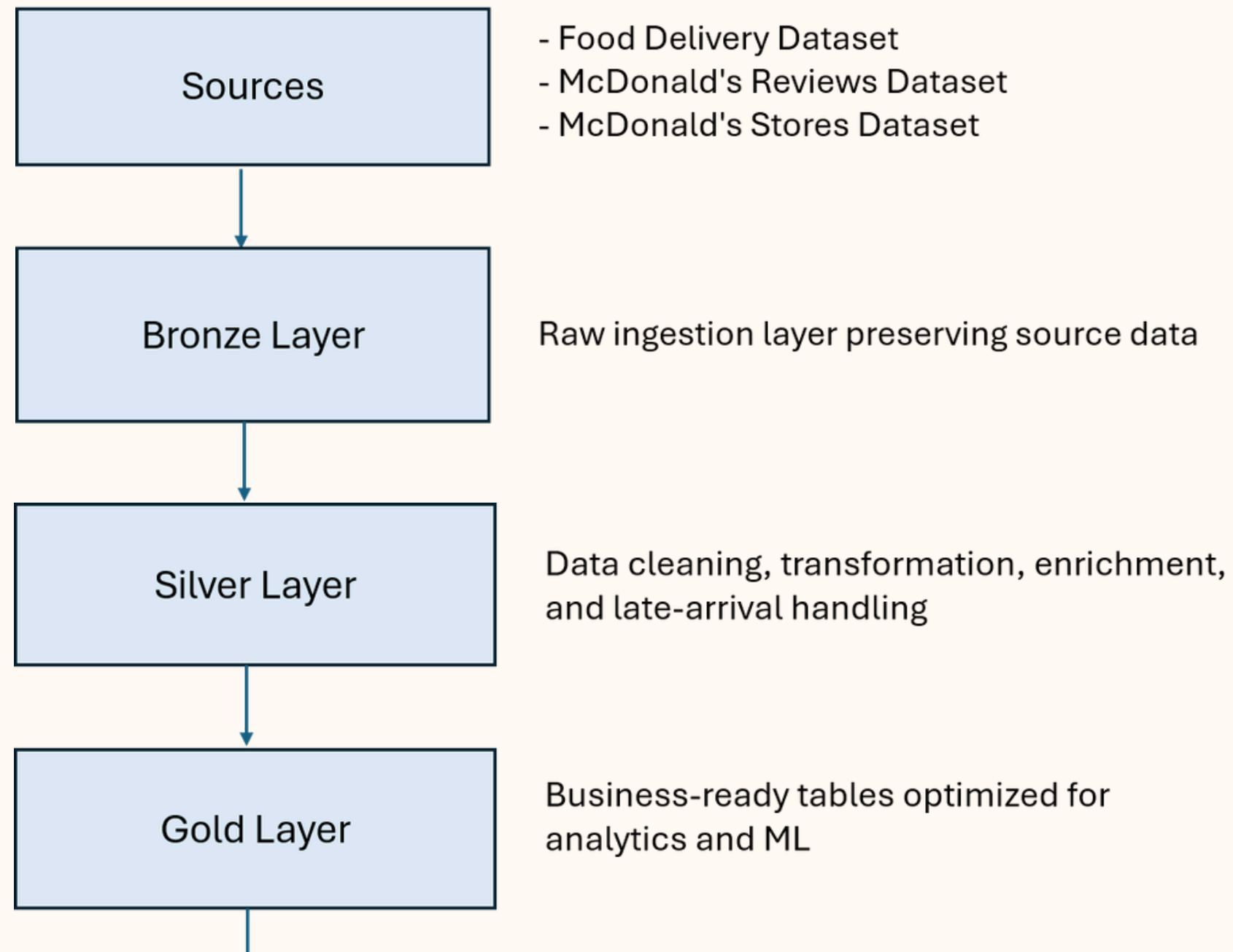
McDonald's Stores Data:https://www.kaggle.com/datasets/forveryou/mcdonalds-stores-data?utm_source=chatgpt.com





Architecture

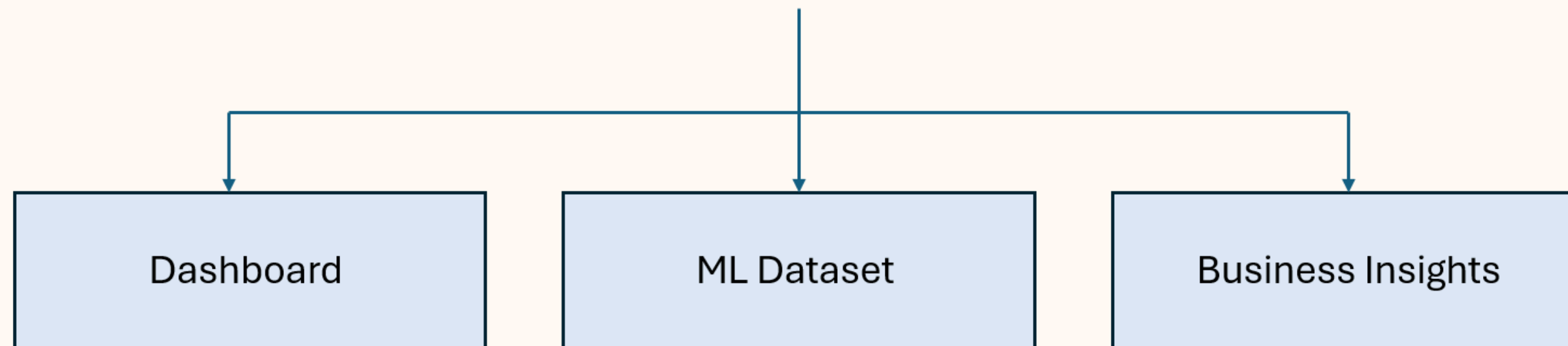
High-level Architecture





Architecture

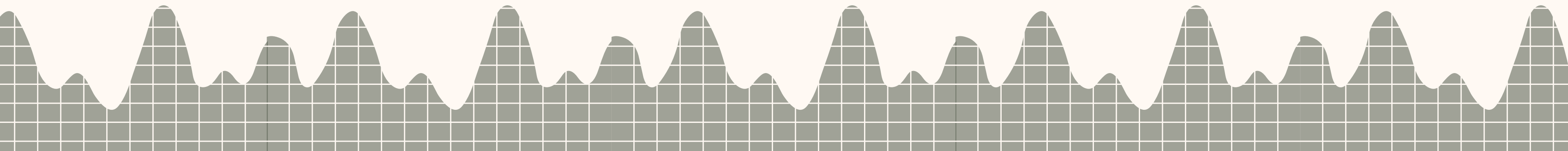
High-level Architecture



Business performance monitoring and KPI visualization.

Feature tables for predictive modeling.

Data-driven decisions and branch optimization.





Bronze Tables

Bronze-delivery events

Column Name	Data Type
order_id	string
restaurant_id	string
order_time	timestamp
delivery_time	int
order_value	float
traffic_condition	string
ingestion_time	timestamp

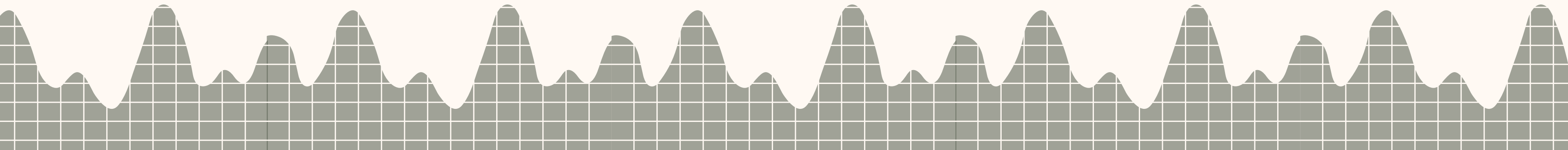
Bronze-reviews

Column Name	Data Type
reviewer_id	int
store_name	string
rating	int
review_time	timestamp
review	text
rating_count	int
ingestion_time	timestamp

Bronze-store metadata

Column Name	Data Type
storeid	string
name	string
country	string
city	string
longitude	string
latitude	float
file_date	timestamp

Bronze Layer Principles: Raw data only,Original source format preserved,Metadata retained for traceability



Silver Tables

Silver_delivery_metrics

Column Name	Transformation
order_id	Preserve source key
restaurant_id	Standardized branch reference
order_time	Standardized timestamp format
delivery_duration	Renamed from delivery_time for clarity
order_value	Validated numeric values
traffic_condition	Standardized categories
ingestion_time	Metadata retention

Silver_customer_reviews

Column Name	Transformation
reviewer_id	Preserve source key
store_name	Standardized branch naming
rating	Converted to numeric format
event_time	Renamed from review_time
clean_review_text	Derived from review after text cleaning and null handling
rating_count	Standardized review count format
ingestion_time	Supports late-arrival processing
sentiment_category	Derived from clean_review_text sentiment analysis

Silver_store_details

Column Name	Transformation
storeid	Standardized store identifier
store_name	Renamed from name for consistency
country	Geographic normalization
city	Standardized location naming
longitude	Coordinate validation and formatting
latitude	Coordinate validation and formatting
file_date	Metadata retention
store_key	Generated surrogate key for warehouse joins

Silver Layer Principles: cleans and enriches raw data to create analysis-ready datasets while supporting joins, late-arrival handling, and downstream processing.

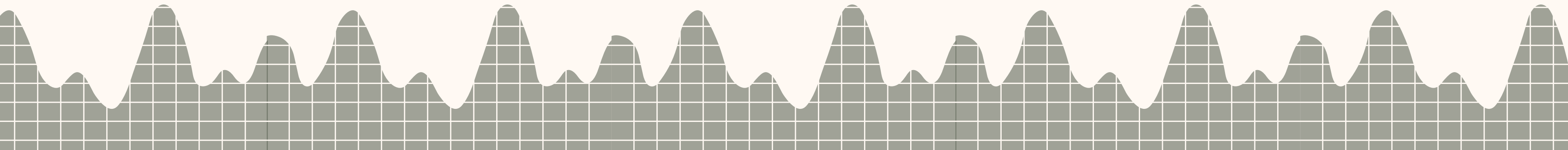


Gold Tables

Gold Fact Tables

Fact Table	Main Fields	Business Purpose
fact_orders	order_key, store_key, order_time, order_value	Tracks revenue and sales activity
fact_delivery_performance	order_key, store_key, delivery_duration, traffic_condition,is_late_delivery	Measures service efficiency
fact_customer_reviews	review_key, store_key, rating, event_time	Captures customer satisfaction

Gold fact tables transform cleaned Silver data into focused business event tables that support profitability analysis, dashboard metrics, and ML feature creation.





fact_orders

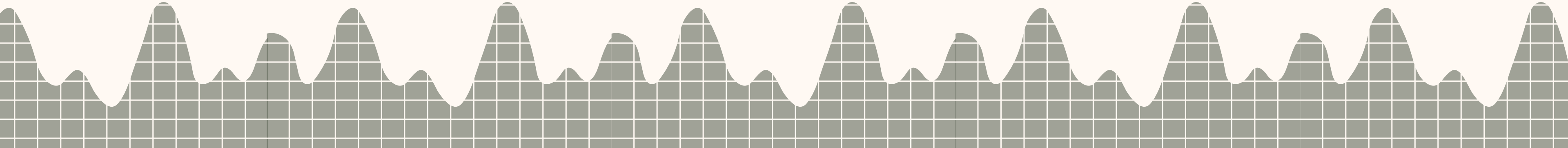
Column Name	Data Type
order_key	string
store_key	string
order_time	timestamp
order_value	float

fact_delivery_performance

Column Name	Data Type
order_key	string
order_key	string
order_key	int
order_key	string
order_key	boolean

fact_customer_reviews

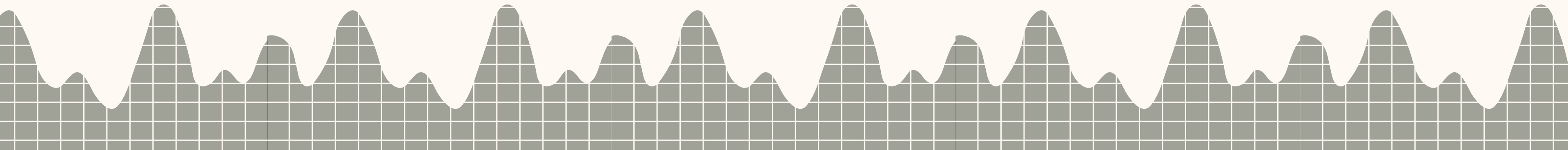
Column Name	Data Type
review_key	string
store_key	string
rating	int
event_time	timestamp



Gold Aggregated Tables

Aggregated Table	Main Fields	Business Purpose
agg_branch_profitability	store_key, total_revenue, avg_order_value, profitability_score	Measures branch profitability and financial performance
agg_delivery_performance	store_key, avg_delivery_duration, late_delivery_rate, traffic_index	Tracks delivery efficiency and operational bottlenecks
agg_customer_experience	store_key, avg_rating, negative_review_rate, sentiment_score	Measures customer satisfaction and service quality
agg_branch_summary	store_key, total_orders, avg_rating, profitability_score, late_delivery_rate	Combines operational and customer KPIs at branch level

Gold aggregated tables summarize operational and customer events into branch-level KPIs used for dashboards, profitability analysis, and predictive modeling.





agg_branch_profitability

Column Name	Data Type
store_key	string
total_revenue	float
avg_order_value	float
profitability_score	float

agg_delivery_performance

Column Name	Data Type
store_key	string
avg_delivery_duration	float
late_delivery_rate	float
traffic_index	float



agg_customer_experience

Column Name	Data Type
store_key	string
avg_rating	float
negative_review_rate	float
sentiment_score	float

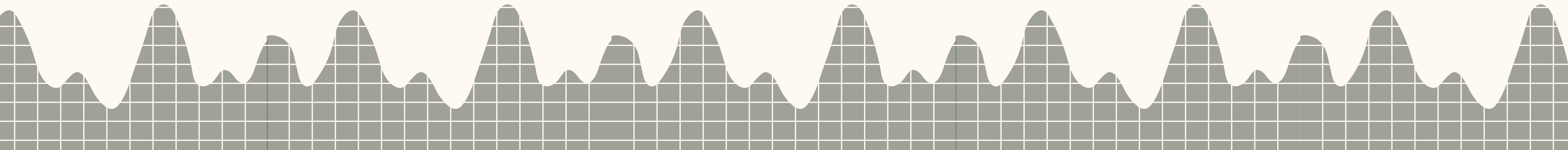
agg_branch_summary

Column Name	Data Type
store_key	string
total_orders	int
avg_rating	float
profitability_score	float
late_delivery_rate	float

Gold Dimension Tables

Dimension Table	Main Fields	Business Purpose
dim_store	store_key, storeid, store_name, city, country, latitude, longitude	Provides branch identity, location, and geographic context
dim_date	date_key, date, day, month, year, weekday	Enables time-based analysis of orders, reviews, and performance
dim_traffic_condition	traffic_key, traffic_condition	Supports analysis of how traffic affects delivery performance
dim_customer_feedback	feedback_key, sentiment_category, rating_group	Classifies reviews into satisfaction and sentiment categories

Gold dimension tables provide descriptive business context used to filter, segment, and join fact tables for dashboard reporting and ML feature creation.





dim_store

Column Name	Data Type
store_key	string
storeid	string
store_name	string
city	string
country	string
latitude	float
longitude	float

dim_date

Column Name	Data Type
date_key	int
date	date
day	int
month	int
year	int
weekday	string



dim_traffic_condition

Column Name	Data Type
traffic_key	int
traffic_condition	string

dim_customer_feedback

Column Name	Data Type
feedback_key	int
sentiment_category	string
rating_group	string



SCD Type 2

SCD Type 2: Store Dimension History

Purpose:

Track historical changes in store attributes without overwriting previous records.

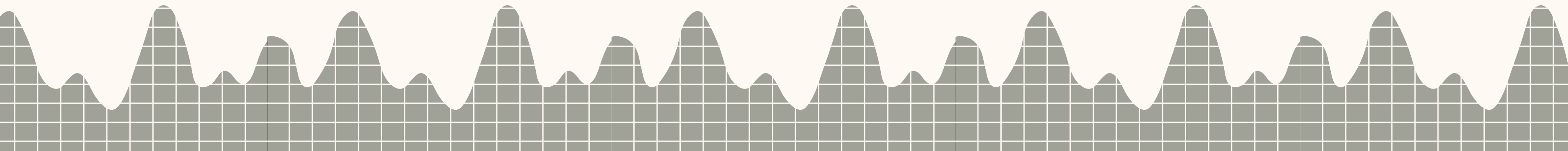
In this project, dim_store is managed as a Type 2 Slowly Changing Dimension because store attributes may change over time, such as location details, operating region, or available services.

If McDonald's updates a store's attributes, overwriting the old record would damage historical analysis.

Example:

A branch had Drive-Thru = No during January–March, then added Drive-Thru in April.

Without SCD Type 2, old orders would incorrectly appear as if they happened when the branch already had Drive-Thru.





SCD Type 2

Field	Role
store_key	Surrogate key used in fact tables
storeid	Original business/store identifier
store_name	Branch name
city	Store city
country	Store country
services	Available branch services
valid_from	Start date of this version
valid_to	End date of this version
is_current	Marks the active version

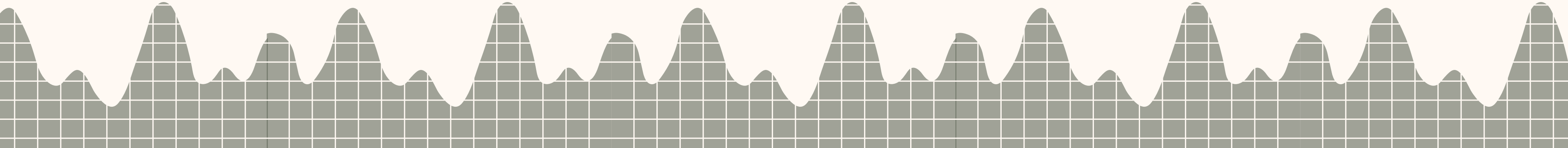


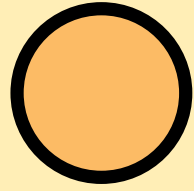
SCD Type 2

store_key	storeid	store_name	city	country	services	valid_from	valid_to	is_current
1001	5021	McDonald's Shanghai Central	Shanghai	China	Dine-in, Delivery	01/01/2024	15/04/2025	FALSE
1187	5021	McDonald's Shanghai Central	Shanghai	China	Dine-in, Delivery, Drive-Thru	15/04/2025	NULL	TRUE
1002	3340	McDonald's Houston Main	Houston	United States	Dine-in, Drive-Thru	01/01/2024	01/02/2025	FALSE
1224	3340	McDonald's Houston Main	Houston	United States	Delivery, Dine-in, Drive-Thru	01/02/2025	NULL	TRUE
store_key	storeid	store_name	city	country	services	valid_from	valid_to	is_current
1001	5021	McDonald's Shanghai Central	Shanghai	China	Dine-in, Delivery	01/01/2024	15/04/2025	FALSE
1187	5021	McDonald's Shanghai Central	Shanghai	China	Dine-in, Delivery, Drive-Thru	15/04/2025	NULL	TRUE
1002	3340	McDonald's Houston Main	Houston	United States	Dine-in, Drive-Thru	01/01/2024	01/02/2025	FALSE
1224	3340	McDonald's Houston Main	Houston	United States	Delivery, Dine-in, Drive-Thru	01/02/2025	NULL	TRUE

How the update works
When a tracked attribute changes:

- The old record is closed by setting valid_to.
- is_current becomes False.
- A new record is inserted with a new store_key.
- The new record gets valid_to = NULL and is_current = True.





Executive Profitability



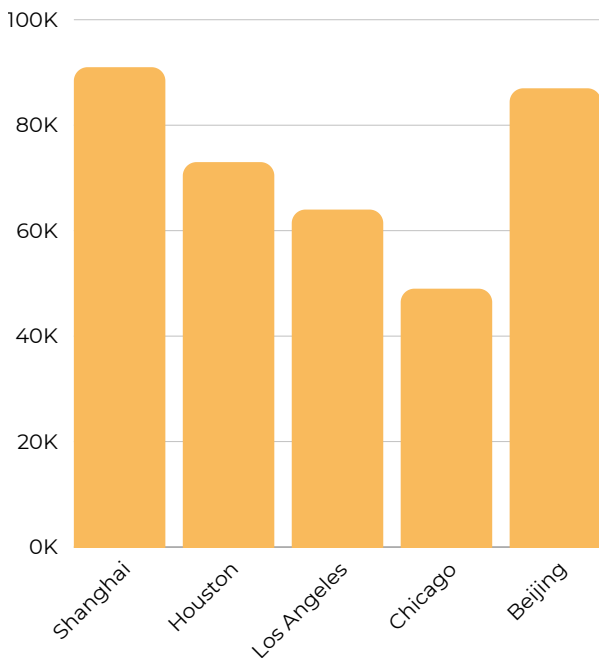
Total Revenue

Avg Order Value

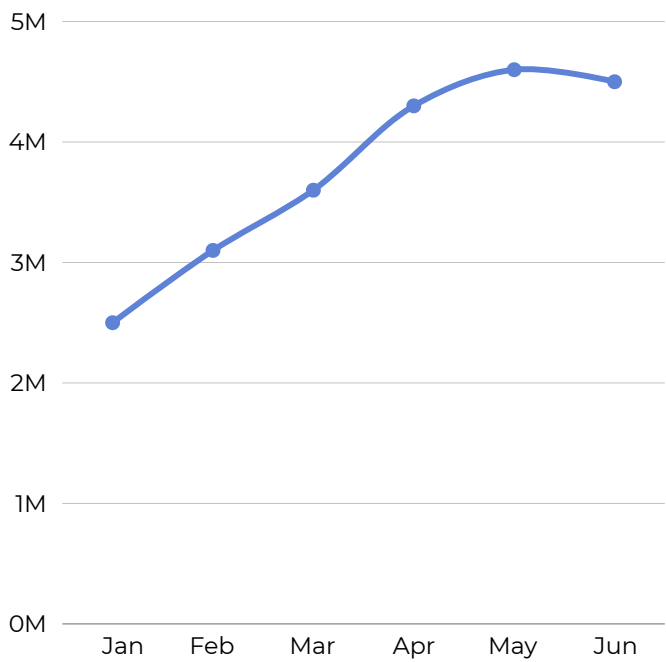
Profitability Score

Underperforming
Branches

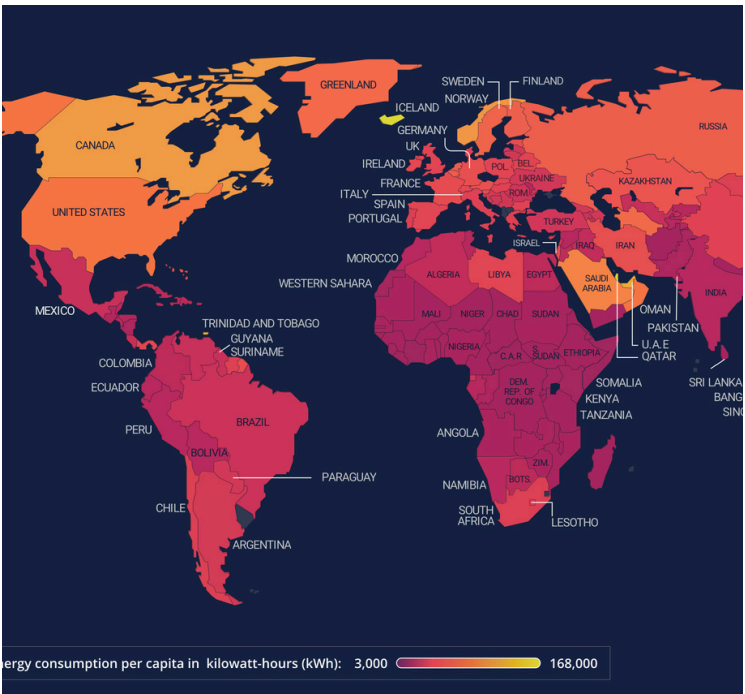
Executive Profitability

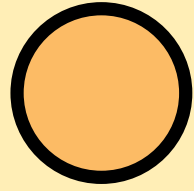


Revenue Trend

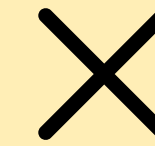


Profitability by Region





Operations & Delivery



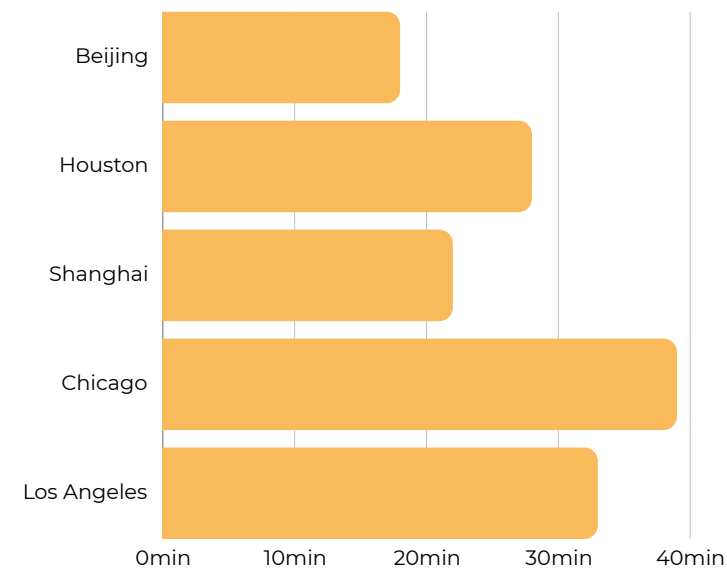
Avg Delivery
Duration

Late Delivery
Rate

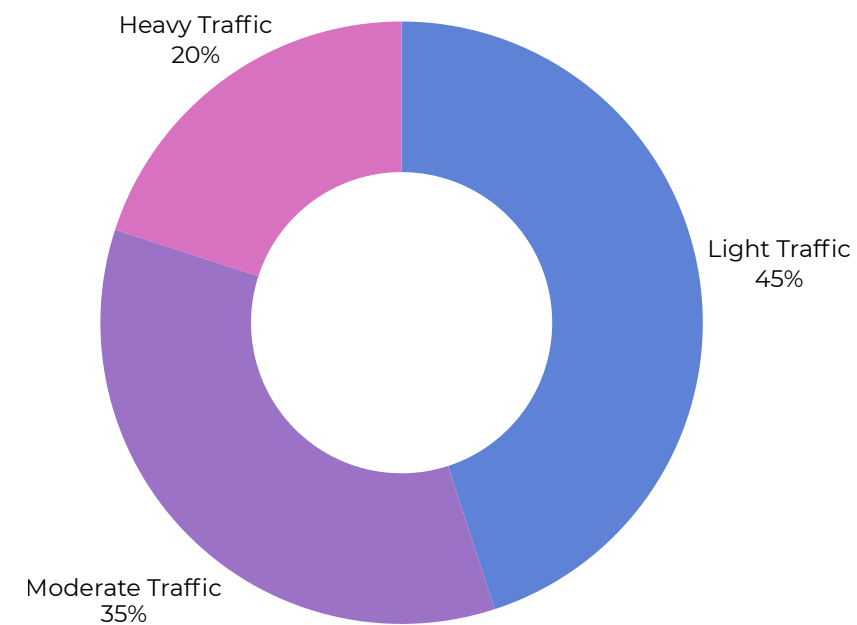
Orders
Processed

Traffic Impact
Score

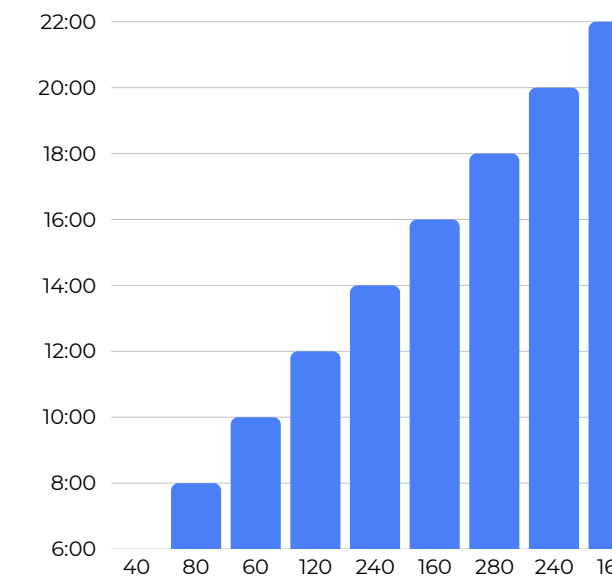
Average Delivery
Duration by Branch



Traffic Condition Distribution



Orders Volume
Throughout the Day

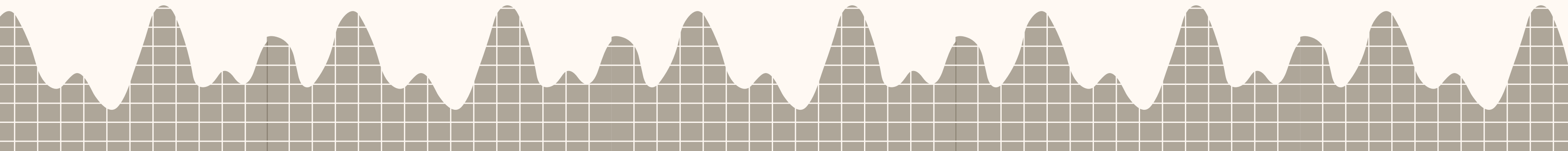




ML Business Problem

Machine Learning question

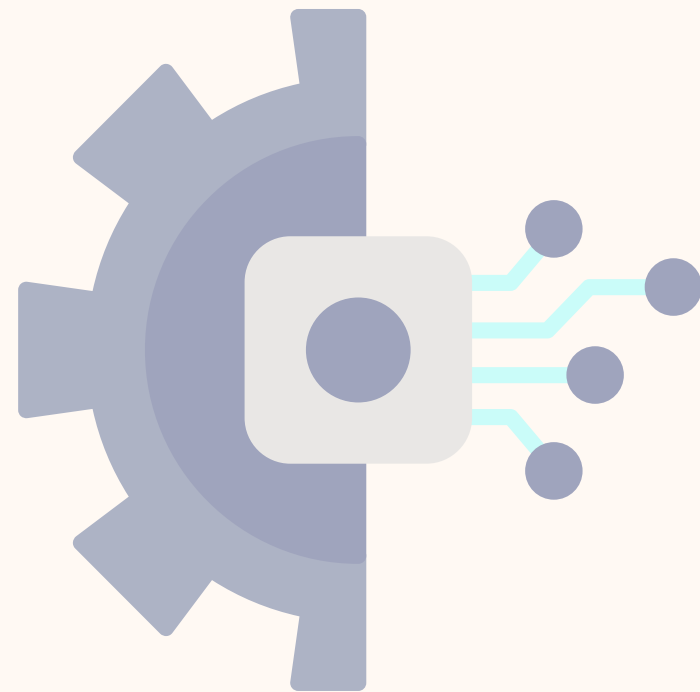
Can we predict whether a McDonald's branch is likely to become underperforming using operational, customer experience, and profitability indicators?





ML Business Problem

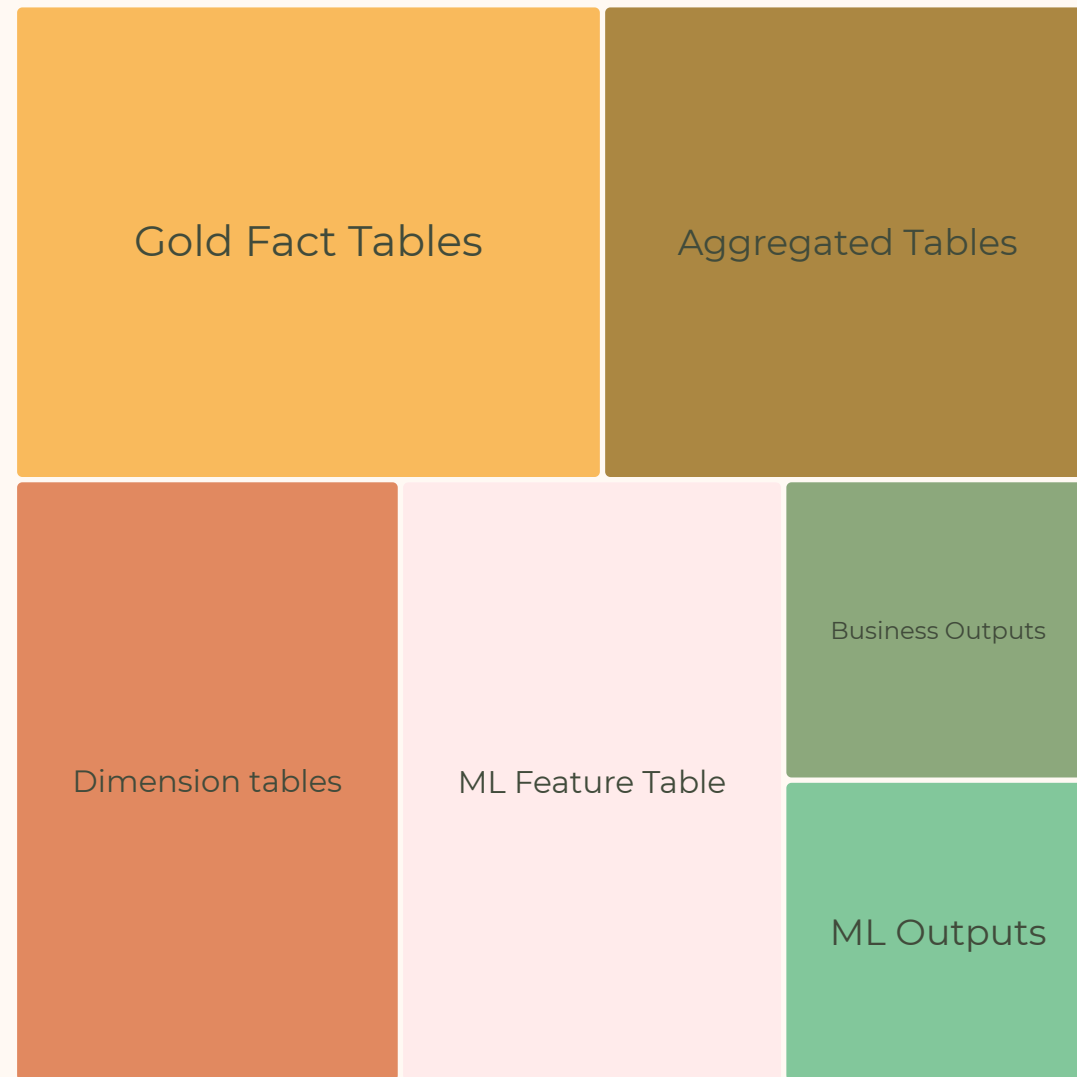
ML Feature Table



Feature Name	Role	Gold Source Field
store_key	Identifier	dim_store.store_key
avg_delivery_duration	Feature (X)	agg_delivery_performance.avg_delivery_duration
late_delivery_rate	Feature (X)	agg_delivery_performance.late_delivery_rate
avg_rating	Feature (X)	agg_customer_experience.avg_rating
negative_review_rate	Feature (X)	agg_customer_experience.negative_review_rate
sentiment_score	Feature (X)	agg_customer_experience.s entiment_score
total_revenue	Feature (X)	agg_branch_profitability.to tal_revenue
total_orders	Feature (X)	agg_branch_summary.total _orders
profitability_score	Feature (X)	agg_branch_profitability.pr ofitability_score
underperforming_branch	Target (Y)	Derived label generated from business KPI thresholds



Gold Layer → Dashboard & ML Flow



Gold tables are transformed into analytical assets by aggregating operational, customer, and profitability metrics into business-ready datasets. These datasets support business intelligence dashboards for monitoring branch performance and also serve as the foundation for machine learning workflows used to identify and predict underperforming branches.



Conclusion

Conclusion & Business Value

Key Achievements

- Integrated streaming, late-arrival, and batch data sources
- Built Bronze, Silver, and Gold warehouse layers
- Implemented historical tracking using SCD Type 2
- Designed business intelligence dashboards
- Created ML-ready datasets for branch prediction

Business Impact

- Improved visibility into branch profitability drivers
- Faster identification of operational bottlenecks
- Better customer experience monitoring
- Early detection of underperforming branches
- Support for proactive decision-making

By combining data engineering, analytics, and machine learning, the platform enables McDonald's to move from reactive reporting toward proactive branch optimization.

